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Laser Diagnostic Study of the Mechanism of a Periodic Combustion Instability in a Gas Turbine Model Combustor

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Abstract. To investigate the mechanisms leading to sustained thermoacoustic oscillations in swirl flames, a gas turbine model combustor was equipped with an optically accessible combustion chamber allowing the application of various laser techniques. The flame investigated was a swirled CH_4/air diffusion flame (thermal power 10 kW, global equivalence ratio $\phi = 0.75$) at atmospheric pressure which exhibited self-excited thermoacoustic oscillations at a frequency of 290 Hz. In separate experiments, the flow velocities were measured by laser Doppler velocimetry, the flame structures and heat release rates by planar laser-induced fluorescence of CH and by OH chemiluminescence, and the joint probability density functions of the major species concentrations, mixture fraction, and temperature by laser Raman scattering. All measurements were performed in a phase-locked mode, i.e., triggered with respect to the oscillating pressure level measured by a microphone. The results revealed large periodic variations of all measured quantities and showed that the heat release rate was correlated with the degree of mixing of hot products with unburned fuel/air mixtures before ignition. The thermal expansion of the reacting gases had, in turn, a strong influence on the flow field and induced a periodic motion of the inner and outer recirculation zones. The combination of all results yielded a deeper understanding of the events sustaining the oscillations in the flame under investigation. The results also represent a data base that can be used for the validation and improvement of CFD codes.

Key words: thermoacoustic, swirl flames, laser diagnostic, validation measurements

1. Introduction

Thermoacoustic instabilities are often encountered in modern gas turbine (GT) combustors and pose a severe problem for further NO_x reduction by applying ultra-lean combustion concepts [1–4]. Periodic pressure oscillations are driven by the combustion process, which in turn is subject to pulsations that can be sustained by a resonant feedback mechanism coupling pressure and heat release [5, 6]. The physical and chemical mechanisms driving the instabilities are based on a complex interaction between combustor geometry, pressure, flow field, mixing, chemical reactions, and heat release, and are not understood well enough yet. Because the

problems could not be solved fundamentally so far, combustion instabilities have been reduced or eliminated in industrial burners by passive or active control mechanisms [7, 8]. On the other hand, large efforts are currently being undertaken in the development of numerical simulation tools in order to predict unsteady combustion behavior so that improved GT combustors can be designed with their help. However, a deeper understanding of the complex interactions involved in combustion instabilities is now and in the near future based on experimental research using advanced measuring techniques with high temporal and spatial resolution.

The German Aerospace Center (DLR) has established a lab scale "standard burner" for swirling CH_4/air diffusion flames at atmospheric pressure that is close to a practical GT combustor but with large windows for optical measurements. Several laser techniques have been applied in order to gain insight into processes leading to the instabilities and to provide an experimental data base which can be used for the validation of numerical simulations. The investigated flame had a thermal power of 10 kW, a global equivalence ratio of $\phi = 0.75$, and it exhibited thermoacoustic oscillations with a dominant frequency of 290 Hz which were recorded by microphones. This instability was very well reproducible enabling the comparison of experiments that were performed at different times. Laser Doppler velocimetry was used to measure the flow velocities simultaneously with the pressure signal from a microphone in order to correlate velocities and pressure level. Planar laser-induced fluorescence (PLIF) was applied to measure two-dimensional CH structures in the flame and laser Raman scattering to determine the major species concentrations, temperature, and mixture fraction. These measurements were triggered with respect to the phase of the pressure pulsations in the combustor plenum, enabling phase-resolved measurements. In addition, OH^* chemiluminescence was detected as a measure for the heat release rate.

Although the instantaneous flame behavior was dominated by turbulent fluctuations, measurements of quantities averaged at fixed phase angles revealed significant periodic variations of the flame, especially in the near-field. The heat release rate varied by more than $\pm 20\%$ during an oscillation cycle and was accompanied by pressure variations. At the inlet, the mean flow velocities also varied by more than 20% in magnitude. The corresponding distributions of mixture fraction, species concentrations, and temperature provided an insight into the chain of events during an oscillation period.

2. Experimental

2.1. Combustor and flame

The gas turbine model combustor is schematically shown in Figure 1. Co-swirling dry air at room temperature and atmospheric pressure is supplied to the flame through a central nozzle (diameter 15 mm) and an annular nozzle (i.d. 17 mm, o.d. 25 mm contoured to an o.d. of 40 mm). Non-swirling CH_4 is fed through 72

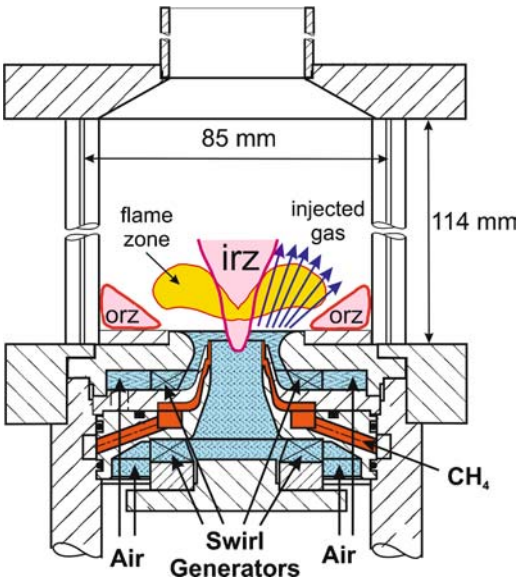

Figure 1. Nozzle and combustion chamber of the gas turbine model combustor. Also indicated are the flame zone, the inner and outer recirculation zones IRZ and ORZ, respectively, and the direction of the inlet flow of the injected gases.

channels (0.5 mm × 0.5 mm) forming a ring between the air nozzles. The exit plane of the fuel and central air nozzles is located 4.5 mm below the exit plane of the outer air nozzle which defines the height $h = 0$. The burner is a modified version of a gas turbine combustor with an air blast nozzle for liquid fuels [9, 10]. The combustion chamber consists of four quartz plates held by four steel rods (diameter 10 mm) in the corners allowing very good optical access to the flame. The chamber has a square section of 85 mm × 85 mm and a height of 114 mm. A conical top plate with a central exhaust tube (diameter 40 mm, length 50 mm) forms the exit. The flame investigated was operated with 218.0 standard liters per minute (slpm) air and 17.2 slpm CH_4, corresponding to a thermal power of 10.3 kW and a global equivalence ratio of $\phi = 0.75$. Under these conditions the flame exhibited periodic acoustic oscillations with a frequency of 290 Hz. The pressure oscillations were measured by two microphones, one mounted in the combustor plenum about 70 mm below the plane with $h = 0$ and the other at one steel rod in the corner of the combustion chamber at $h = 20$ mm.

2.2. Measuring techniques

The axial and radial velocity components were measured simultaneously with the pressure signal from the plenum to be able to correlate the velocity data and the pressure fluctuation. The LDV system consisted of a bidirectional optic component from DISA/DANTEC (DISA 55X) and an Ar^+laser (INNOVA, Coherent) using

two different laser wavelengths ($\lambda_{laser} = 488$ and 514.5 nm). A frequency shift of 40 MHz was applied for both directions. Using focusing lenses with $f = 300$ mm, the probe volumes were about 60 μm in diameter and 1.0 mm in length. SiO_2 particles with a diameter of 0.8 μm were seeded as scatterers into the air flow. For the simultaneous detection and analysis of the photomultiplier signals, two Dantec burst spectrum analyzers were used (BSA enhanced, 57N20 and 57N35). At each measuring location, typically 10,000–15,000 velocity data were recorded with the center frequency being adapted at each measuring position. To correlate the velocity signals to the oscillating pressure level, the positive slope of the pressure signal has been recorded via a third BSA (57N35), thus logging the pressure periodicity. In a post processing step the velocity data were sorted according to the pressure fluctuation into 36 intervals per period (resulting in a $\pm 5^\circ$ resolution) by comparing the arrival time of each velocity signal with the corresponding time of the pressure signal. Phase-resolved velocity measurements were performed at $h = 1$ and 5 mm, within radial intervals of $\Delta r = 1$ or 2 mm. At heights $h > 5$ mm, measurements were performed without phase-resolution. The uncertainty of the measurements is less than 1% for the mean values.

The flame structures were visualized by planar laser-induced fluorescence of CH. The tunable frequency-doubled laser radiation of a pulsed optical parametric oscillator (Spectra Physics GCR 290 and MOPO 730) was formed to a light sheet of approximately 50 mm height and irradiated vertically into the flame. The excited fluorescence was collected at 90° by a camera lens ($f = 50$ mm, $f/0.95$, Canon) and, after spectral filtering, detected by an intensified CCD camera (Roper Scientific). The laser pulse duration was 5 ns, the temporal detection gate of the image intensifier was 200 ns. The CH radicals were excited on the $Q_1(7)$ line of the $B^2\Sigma - X^2\Pi$ ($\nu' = 0$, $\nu'' = 0$) band at 390.3 nm [11]. For the suppression of laser scattered light and background radiation, a filter combination of a KV418 and a short pass filter KP450 was used in front of the camera and only the fluorescence from the A state around $\lambda \approx 430$ nm was detected. This state is efficiently populated by collision-induced electronic energy transfer from the B to the A electronic level [12, 13]. In flames, CH radicals are formed at high temperatures on the fuel-rich side of the reaction zone and have a short lifetime ($\approx$10 ns). Thus, high CH concentrations can be interpreted as a marker for the fuel-consumption layer of the reaction zone and, with some restrictions, as a qualitative measure for the heat release rate [14, 15]. The chemiluminescence of OH^* was measured by an intensified CCD camera equipped with an UV achromatic lens ($f = 100$ mm, B. Halle Nachf.) and an interference filter centered at $\lambda = 312$ nm. OH^* is an intermediate species in CH_4/air combustion [16] and is regarded as a good measure for the heat release rate [17, 18].

For the pointwise quantitative measurement of the major species concentrations (O_2, N_2, CH_4, H_2, CO, CO_2, H_2O) and the temperature, laser Raman scattering was applied separately from the other measurements. The radiation of a flashlamp-pumped dye laser (Candela LFDL 20, wavelength $\lambda = 489$ nm, pulse energy

$E_p \approx 3$ J, pulse duration $\tau_p \approx 3\,\mu$s) was focused into the combustion chamber and the Raman scattering emitted from the measuring volume (length 0.6 mm, diameter 0.6 mm) was collected by an achromatic lens and relayed to the entrance slit of a spectrograph. The dispersed and spatially separated signals from the different species were detected by photomultiplier tubes in the exit plane of the spectrograph. The species number densities were calculated from these signals using calibration measurements and the temperature was deduced from the total number density via the ideal gas law [19, 20]. The simultaneous detection of all major species with each laser pulse also enabled the determination of the instantaneous mixture fraction [21]. At each measuring location within a pattern of roughly 50 points, covering the heights $h = 5$, 10, 20, 30, and 40 mm, 400 single-pulse measurements were performed from which the joint probability density functions (PDFs) were established. At heights $h \geq 40$ mm phase-dependent variations were smaller than the measurement accuracy and no phase-locking was applied for measurements above $h = 40$ mm. The measurement uncertainty due to systematic errors is typically 3–4% for the temperature and mixture fraction, and 3–5% for the species mole fractions of O_2, CH_4, CO_2, and H_2O. The single-pulse uncertainties, including statistical errors due to photon statistics, are 4–5% for temperature and mixture fraction and typically 5–10% for the mole fractions, depending on the number density [22, 23].

All pulsed measurements were performed in a phase-locked mode, i.e., triggered with respect to the oscillating pressure signal measured by the microphone in the combustor plenum [10, 24]. The phase angle $\alpha = 0^\circ$ was defined at the pressure minimum in the plenum and measurements were performed at steps of $45^\circ \pm 5^\circ$. The pressure curve was not exactly sinusoidal but exhibited a slight asymmetry in the form of a slightly faster pressure decrease than increase. However, for simplicity the pressure curve will be treated as a pure sine curve in the following. The signal from the microphone in the combustion chamber was not used for the triggering because it exhibited a poorer signal-to-noise ratio. The results displayed in this paper are phase-resolved mean values, i.e., averaged at distinct phase angles.

3. Results and Discussion

3.1. Acoustic behavior

Figure 2 shows an exemplary acoustic wave train measured by the microphone in the combustor plenum and the corresponding frequency spectrum. The dominant frequency is $f_0 \approx 290$ Hz but there is also a small contribution from the overtone oscillation at 580 Hz. f_0 corresponds to a period of $\tau_0 = 3.44$ ms and, assuming an average temperature of about 1700 K in the combustion chamber resulting in a speed of sound of approximately 800 m/s, a wavelength of 2.75 m. Thus, the wavelength is large compared to the dimensions of the combustion chamber. The pressure variations in the combustor plenum lagged by about 60° relative to the

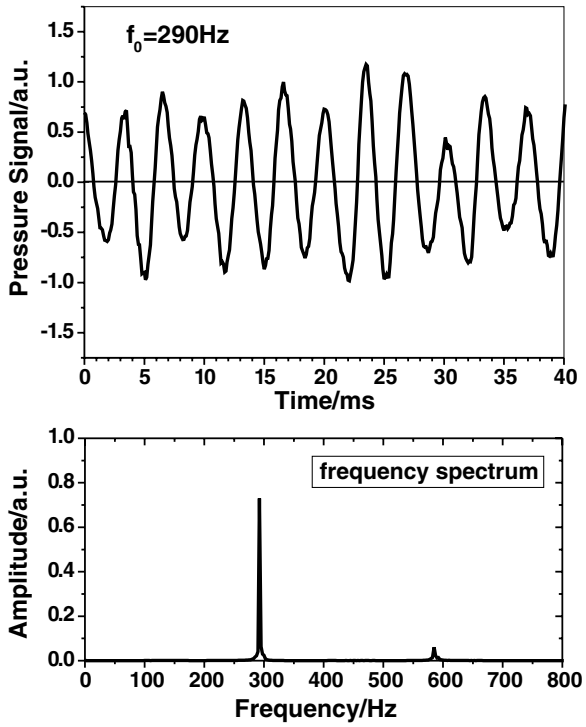

Figure 2. Signal measured by the microphone in the combustor plenum. Upper panel: exemplary wave train. Lower panel: frequency spectrum from the Fourier transformed microphone signal.

pressure measured in the corner of the combustion chamber. This phase shift may be explained by the time the pressure wave needs to travel from the combustion chamber to the position of the microphone in the plenum and by the acoustic behavior of the system [24].

3.2. Velocities

Figure 3 shows radial profiles of the mean axial and radial velocity components, u and v, respectively, at a height $h = 1$ mm for four different phase angles α with a sampling window of $\pm 5°$. Near the flame axis, a well-pronounced inner recirculation zone (IRZ) can be seen with $u_{\text{mean}}(r = 0)$ varying between -9.0 m/s at $\alpha = 315°$ and -1.2 m/s at $\alpha = 135°$. These large phase-dependent changes of u_{mean} show, that the IRZ is moving up and down and that it extends down into the central nozzles despite of the upstream directed air flow. The highest positive axial velocities are found in the radial region $r \approx 5$–15 mm where air and CH_4 are injected into the chamber. Here, the maximum of u varies from $u_{\text{mean}} = 12.3$ m/s at $\alpha = 225°$ to 16.2 m/s at $\alpha = 45°$. The radial velocity component v reaches mean

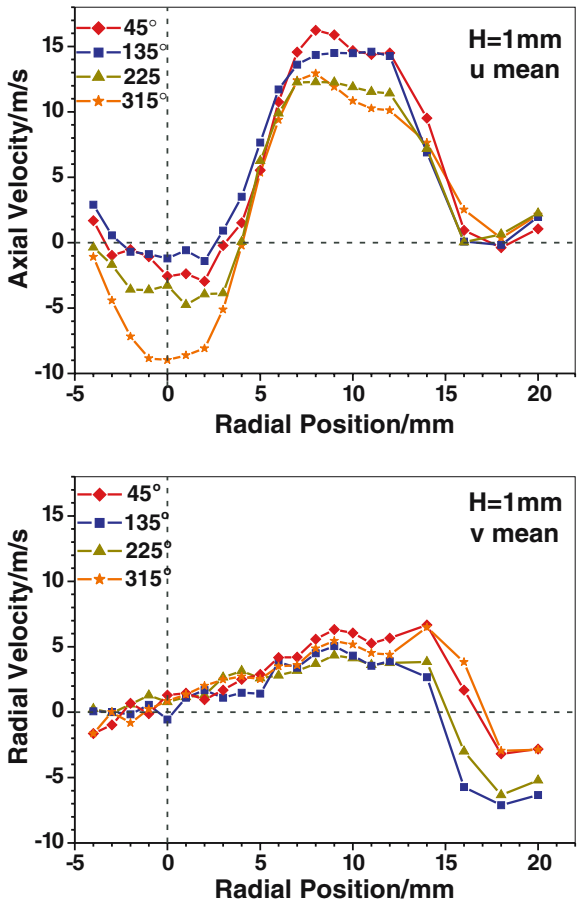

Figure 3. Radial profiles of the mean axial and radial velocity components at $h = 1$ mm for four different phase angles.

values of up to 6.3 m/s in the radial region where the fresh gases are injected and its α-dependent changes are not generally in phase with u_{mean}. Further, the profiles prove the existence of an outer recirculation zone (ORZ) for $r > 15$ mm with negative values of v (directed towards the axis). Here, v_{mean} exhibits a significant variation with α, e.g., at $r = 18$ mm from $v_{\text{mean}} = -3.0$ m/s at $\alpha = 315°$ to -7.1 m/s at $\alpha = 135°$. Note that the variations of v_{mean} in the ORZ are out of phase by $\approx 180°$ compared to the variations of u_{mean} in the IRZ, i.e., when the IRZ becomes stronger and moves down, the ORZ becomes weaker and moves outward and vice versa.

To illustrate the downstream development of the flow field within up to $h = 30$ mm (the region of the flame zone), Figure 4 shows a vectorplot of the combined mean u and v components averaged over all phases. It shows that the IRZ exhibits roughly the shape of a cone with an opening angle of approximately 16°. The

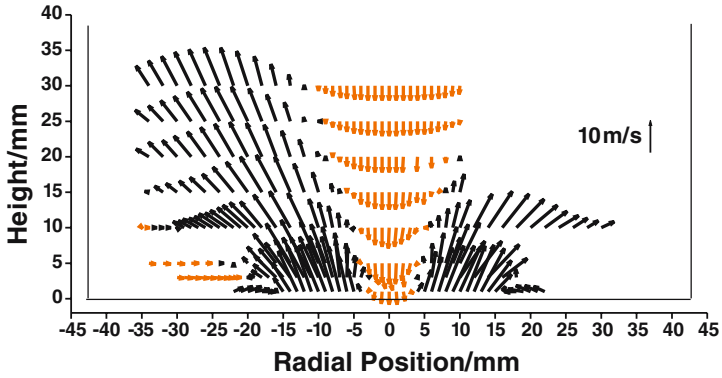

Figure 4. Vectorplot of the combined mean $u - v$ components averaged over all phase angles. Negative axial velocities are displayed red (gray), indicating the recirculation zones. The lines in the outer region indicate the boundaries of the combustion chamber.

surface of this cone corresponds to $u_{\text{mean}} = 0$ and represents the average location of the shear layer between the IRZ and the inlet flow. It is also seen that the ORZ is restricted to $h \leq 10$ mm. The main stream of the inflow becomes rapidly broader: while the flow is restricted to $r \approx 5$–15 mm at $h = 1$ mm, it extends already from $r \approx 6$ to 29 mm at $h = 10$ mm. With respect to the convective transport of fuel, air, and exhaust gas within the inlet stream, typical velocities are around 10 m/s. Therefore, within an oscillation cycle, the gas is transported roughly 35 mm downstream. The sketch within Figure 1 summarizes the main characteristics of the flow field, i.e., the main stream of injected gases and the recirculation zones, together with the indicated flame zone.

3.3. CH PLIF

Planar images of LIF from CH are displayed in Figure 5 for the four different phase angles discussed before. These images have been accumulated over 500 single shots at each phase angle and the intensities can be regarded as an approximate measure for the mean heat release rate. In single shot exposures (not displayed) thin ($d \approx 0.4$ mm) and strongly wrinkled instantaneous CH structures are seen which represent the reaction zones [25]. In the averaged images, the flame looks conically shaped and rather flat, i.e., it burns quite close to the burner plate, and the heat release takes place in a region with $r \approx 0$–35 mm and $h \approx 2$–25 mm, predominantly between $h = 10$ and 20 mm. The flame is not attached to the fuel nozzle and is thus partially premixed before ignition. It is remarkable that mixing and combustion are already completed at $h \approx 25$ mm, a result that is also confirmed by the Raman measurements. The angle of the cone changes only slightly with phase angle, but the size and location of the flame zone and the total heat release rate q vary considerably over one cycle with a maximum at approximately $\alpha = 135°$ and a minimum at

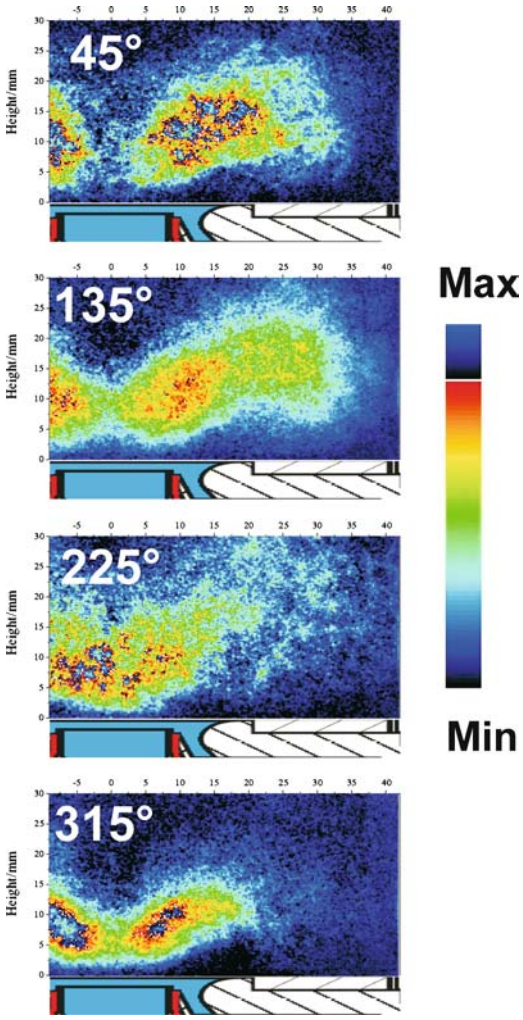

Figure 5. Phase-averaged planar laser-induced fluorescence images of CH for four different phase angles. The vertical light sheet intersects the flame axis and spans the region from $h = 0$ to 30 mm. The wall of the combustion chamber is at the right edge of the image. The nozzle of the combustor is indicated in the lower part of the frames.

approximately $\alpha = 315°$. In particular, the different combustion intensities in the outer radial region ($r \geq 20$ mm) contribute considerably to the variation of q. The phase-resolved measurement of the OH chemiluminescence which is also a good indicator for the heat release rate led to the same result. Compared to the flame under investigation, non-pulsating flames operated at different ϕ and thermal power exhibit a less flat shape. In those flames, reactions and heat release take place predominantly in the shear layer between the injected gases and the IRZ, where hot products from the IRZ mix with unburned fuel/air mixtures and ignite them

[26]. The question arises why in the pulsating flame reactions also occur in the outer radial regions, i.e., within the main stream of injected gases. Here, the Raman measurements help to clarify the mechanism.

3.4. Mixture fraction and temperature

Figure 6 shows radial profiles of the mixture fraction f, the CH_4 mole fraction, and temperature T at $h = 5$ mm for the four different phase angles discussed above. $H = 5$ mm was the lowest measuring height for the Raman measurements because for $h < 5$ mm the optical access was restricted. It can be seen from the CH LIF distributions in Figure 5 that the flame zone starts around this height, more specifically, reactions may occur at $r = 0$–15 mm, but not for $r > 15$ mm. The mean mixture fractions cover a range between $f \approx 0.03$ ($\phi \approx 0.5$) and $f \approx 0.07$ ($\phi \approx 1.3$). These values are quite close to the overall equivalence ratio of $\phi = 0.75$ and demonstrate the fast mixing achieved by this nozzle configuration. The highest f-values are seen at $r \approx 7.5$ mm, i.e., roughly at the position where CH_4 is injected. During an oscillation cycle, considerable variations of f occur with high values at $\alpha = 225°$ and low values at $\alpha = 45°$. The corresponding profiles of CH_4 reach a maximum mean mole fraction of $X \approx 0.09$ around $r = 7.5$–10 mm. Significant variations of CH_4 concentrations are observed on the flame axis with the highest mole fraction at 135° when the reverse flow in the IRZ is slow. A quite surprising result is that the CH_4 distribution extends so far radial outside: in a non-pulsating flame with a very similar flow field, the CH_4 mole fractions were below 0.01 for $r \geq 18$ mm [26]. This is an indication that the pulsations generate an additional mixing process which is already active below $h = 5$ mm. The lower frame of Figure 6 displays the corresponding T-profiles. The largest phase-angle-dependent variations are seen in the IRZ where T varies by roughly 800 K. A comparison with Figure 3 shows that the temperature and the axial velocity on the axis are correlated: at the highest reverse flow velocity, the temperature is maximal. That means, that at $\alpha \approx 315°$ hot gases are transported at a high flow rate into the central air nozzle. It could not be measured how deep the gases penetrate into the nozzle, however, it is clear that they finally mix with air and re-enter the combustion chamber together with the air flow. In the radial region of the downstream flow of the injected gases ($r \approx 5$–21 mm at $h = 5$ mm) the α-dependent temperature variations are on the order of 300 K with a maximum at $\alpha \approx 135°$, i.e., phase-shifted by $\approx 180°$ compared to the variations in the IRZ. The temperature changes in this region are not primarily due to combustion but to transport and mixing processes, i.e., an α-dependent feed of hot gases from the IRZ and the ORZ to the fresh gases. To illustrate this in more detail, the scatterplots of Figure 7 display single-shot results at $h = 5$ mm, $r = 15$ mm for four different phase angles. Each symbol represents the result of a single-shot Raman measurement, the solid line shows for comparison the adiabatic equilibrium state, and the cross indicates the mean value of the single-shot results. It is seen that the samples vary in mixture fraction between $f = 0$ (pure air) and

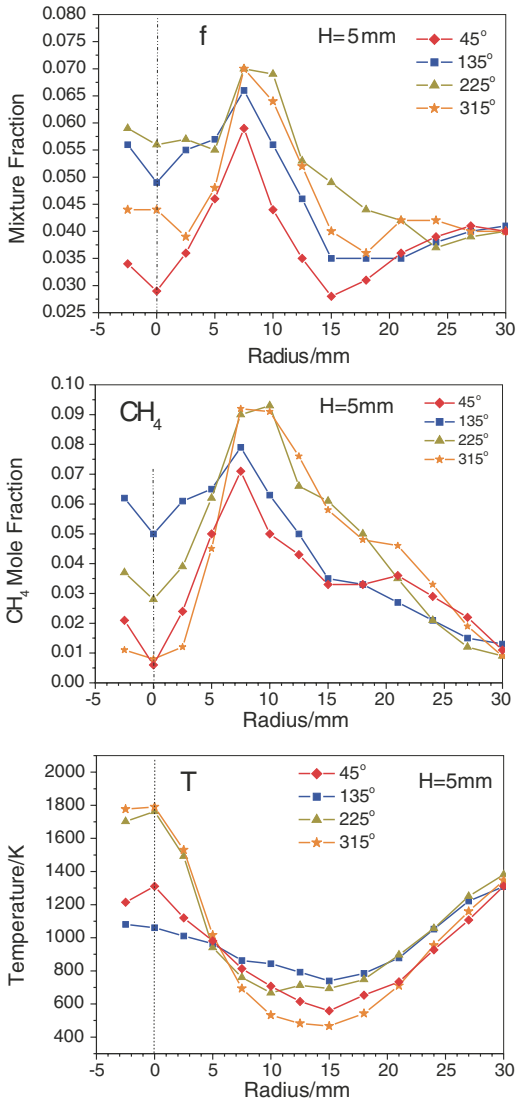

Figure 6. Radial profiles of mean mixture fraction, CH_4 mole fraction, and temperature at $h = 5$ mm at four phase angles. The stoichiometric mixture fraction is $f = 0.055$, the global mixture fraction is $f = 0.0418$.

$f \approx 0.13$ ($\phi \approx 3$). Many probes exhibit temperatures close to room temperature, i.e., they stem from pure mixing of fuel and air without reaction. The samples with elevated temperatures are mixtures of fuel, air, and burnt gas which is convected from the recirculation zones into the inlet flow. The fact that their temperatures are significantly below the adiabatic temperature demonstrates that they have not yet completely reacted due to ignition delay. Only a few samples lying close to

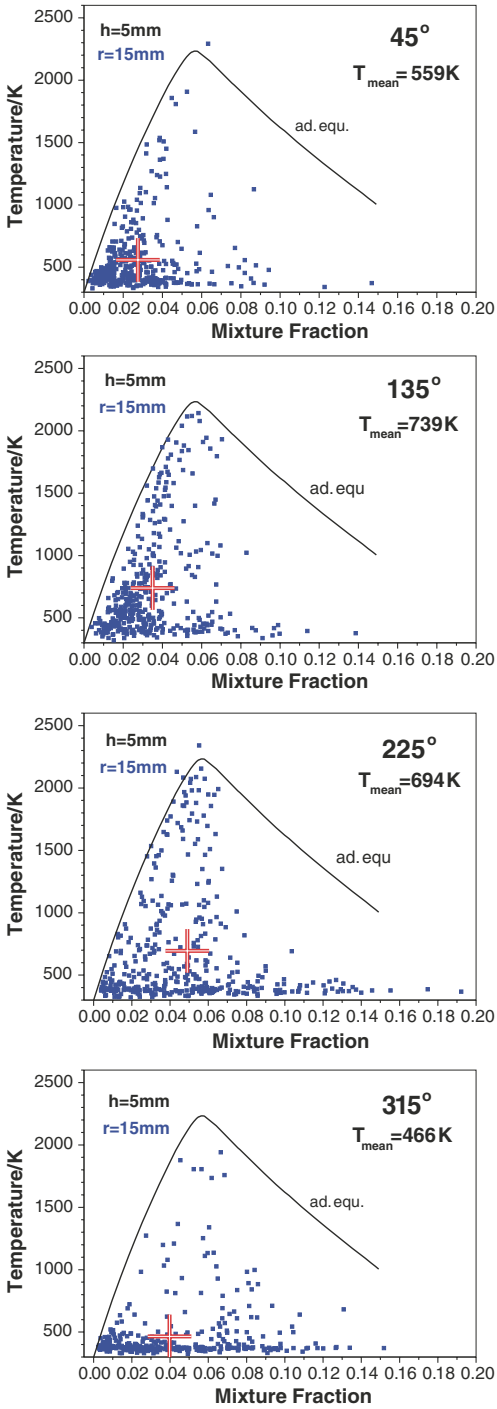

Figure 7. Scatterplots of temperature versus mixture fraction at $h = 5$ mm, $r = 15$ mm for four different phase angles. The crosses indicate the mean values of f and T of the single-shot results.

the adiabatic equilibrium curve can be regarded as completely reacted. Here it cannot be distinguished whether these samples have just reacted or whether they have been convected from the recirculation zones to this location. The comparison of the four scatterplots reveals significant variations during an oscillation cycle: at $\alpha = 315^\circ$ the mean temperature of the samples is at a minimum of 466 K. Later, at $\alpha = 45^\circ$ and 135° the mean temperature increases to 559 and 739 K, respectively. The scatterplots clearly show that this increase is predominantly due to the admixture of burnt gas from the recirculation zones to the fresh gas (around $f \approx 0.025$) and, if at all, only to a small degree to flame reactions. This is an important point for the further discussion of the correlation between heat release rate and temperature variations at $h = 5$ mm.

The temperature variations observed in Figure 6 are consistent with the flow field: at $\alpha \approx 135^\circ$ v_{mean} exhibits the strongest recirculation in the ORZ feeding hot products ($T \approx 1400$ K) to the fresh gases. At the same time, hot gas from the IRZ is mixed into the fresh gas, partly within the shear layer between IRZ and upstream flow, but also to a large extent via a loop through the upper part of the central air nozzle: At $\alpha \approx 315^\circ$ hot gas is strongly sucked into the central nozzle, mixes with air, leaves the nozzle, and finally reaches the plane $h = 5$ mm. The scatterplots of temperature and mixture fraction (not displayed) indicate that the latter occurs at $\alpha \approx 45$–135° and mainly in the radial region $r = 7.5$–12.5 mm. Thus, around 135° the admixture of hot products from the IRZ and ORZ to the unburned fuel/air mixtures reaches a maximum and increases the mean temperature to 700–800 K. This has a pronounced impact on the reactivity: the flammability of the mixtures increases significantly with T and as a result the reaction and heat release rates are promoted.

3.5. Variation of different quantities with phase angle

The phase-dependent changes of pressure and heat release, as well as of temperature, mixture fraction, and CH_4 concentrations at $h = 5$ mm are illustrated in Figure 8. Here, the amplitude of the pressure is enlarged for a better identification. The total heat release rate as determined from the OH chemiluminescence integrated over the entire flame is well correlated with the pressure in the combustion chamber. This behavior is plausible because the heat release causes a thermal expansion and increase of pressure in the combustion chamber which propagates with speed of sound. Because heat release and pressure are nearly in phase, a driving of the pulsation is expected according to the Rayleigh criterion. The temperature displayed in Figure 8 is averaged at $h = 5$ mm over $r = 7.5$ mm to $r = 21$ mm, i.e., the inlet flow of fresh gases. It is seen that the temperature curve is also correlated with the heat release curve. The question arises whether the increase in heat release is the cause for the temperature increase or vice versa. From the results discussed in the preceding paragraph it can be concluded that the temperature increase at $h = 5$ mm is not induced by flame reaction but by convection. On the other hand it is known

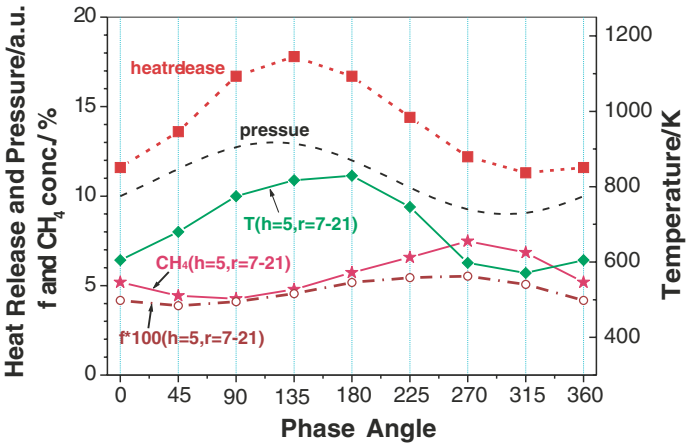

Figure 8. Phase-angle-dependent variations of the heat release rate and pressure in the combustion chamber. Also displayed are the temperature, mixture fraction, and CH_4 concentration (in vol%) averaged over the inlet flow at $h = 5$ mm.

that an admixture of hot burnt gas to cold fresh gas increases the reactivity of the mixture, resulting, e.g., in a reduction of ignition delay times [23]. The observed temperature increase is thus expected to promote the flame reactions occurring a few millimeters further downstream and it seems plausible that this temperature variation of the fresh gas plays an important role for the heat release variations.

To examine whether a variation in fuel concentration or mixture fraction is in phase with the heat release variation, Figure 8 also shows the CH_4 concentrations and f-values averaged over the radial region of the inflow ($r = 7.5$–21 mm). It is seen that the CH_4 concentrations and f are in phase with each other and that both reach a maximum at $\alpha = 270°$. An approximate calculation of molecular fluxes through the plane at $h = 5$ mm, described by Duan et al. for this flame [24], revealed that the CH_4 flux is also at maximum at this phase angle. There is, thus, a periodic change of fuel supply to the flame. The layer with high CH_4 concentrations is convected downstream and reaches the main flame region around $h = 15$ mm at $\alpha \approx 0°$–$45°$. From Figure 8, taking the time delay for convection into account, it can be seen that the heat release rate and CH_4 concentrations are not well correlated. Only for the outer radial regions ($r > 25$ mm at $h = 15$ mm) where the flow velocity is smaller and the time for convective transport longer, the periodic variations of heat release and fuel concentration are approximately in phase. This finding is supported by the radial profiles of mixture fraction at $h = 15$ mm (not displayed) which show the highest f-values for $r \geq 25$ mm at $\alpha = 90°$. An oscillating fuel supply combined with a convective time delay as source for the self-excitation of the pulsations, as reported by Eckstein et al. [27], is therefore for this flame a possible explanation only for the outer radial region. Because the flow velocities were not measured with phase resolution above $h = 5$ mm, a possible influence of periodic

variations of the flow field in the region of the main flame zone around $h = 15$ mm could not be identified.

Although the measurements did not reveal the relative importance of the different sources which might drive the sustained periodic pulsations, the varying amount of admixtures of hot combustion products to the fresh gas is, in the authors' opinion, a main contributor. For the feedback mechanism, the varying swirl intensity plays an important role: The thermal expansion associated with the heat release leads to an acceleration of the flow in axial direction (towards the exhaust) which causes a decrease in swirl intensity (defined as angular momentum flux divided by axial momentum flux) and, thus, a weaker inner recirculation with a minimum at $\alpha \approx 90^\circ$. Figure 9 displays the α-dependent variation of the mean axial velocity u at $h = 1$ mm, $r = 0$, representing the motion of the IRZ, and the variation of the mean radial velocity v at $h = 1$ mm, $r = 18$ mm, representing the ORZ, together with the variations of temperature and heat release. One can see the above mentioned lag between a strong IRZ at $\alpha \approx 315^\circ$ and high temperatures at $h = 5$ mm at $\alpha \approx 135^\circ$–180°. The recirculation in the ORZ, represented by v at $h = 1$ mm, $r = 18$ mm, is approximately in phase with T at $h = 5$ mm, $r = 7$–21 mm and the heat release rate q. Its maximum at $\alpha \approx 135^\circ$ is shifted by $\approx 180^\circ$ compared to the maximum of $u(r = 0)$. That means, that the strengths of the IRZ and ORZ are phase shifted by roughly half a period, but the transport of heat to the inlet flow of fresh gases is in phase at $h = 5$ mm. This pumping mechanism seems to be the main trigger for the reactivity and the heat release which in turn changes the swirl intensity and the flow field.

The sequence of events can be qualitatively summarized as follows: starting at $\alpha \approx 315^\circ$, there is a low heat release rate, a weak ORZ, and a strong recirculation

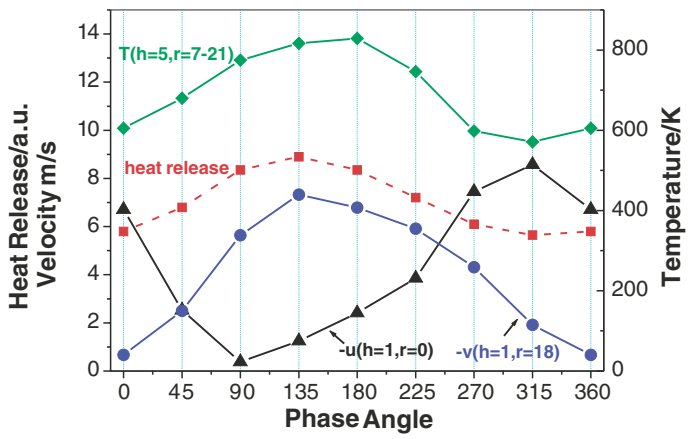

Figure 9. Influence of the phase on the temperature of the inlet flow at $h = 5$ mm, heat release rate, axial velocity u at $h = 1$ mm, $r = 0$, and radial velocity v at $h = 1$ mm, $r = 18$ mm. The velocities are displayed as $-u$ and $-v$.

within the IRZ transporting hot combustion products into the central nozzle. These gases mix with the air flow, re-enter the combustion chamber, and reach the beginning of the flame zone ($h \approx 5$ mm) within the next 2 ms, typically at $\alpha \approx 45°$–$135°$. At the same time, the ORZ becomes stronger and feeds hot combustion products from the outer region into the inlet flow. The growing temperature level in this region increases the reactivity and promotes the heat release rate (maximum at $\alpha \approx 135°$) which in turn results in a thermal expansion with pressure rise and an acceleration of the flow in axial direction. The decrease of swirl intensity following the thermal expansion attenuates the negative pressure gradient along the flame axis and the IRZ decreases in size and strength (minimum at $\alpha \approx 90°$). Now, less hot gas is sucked into the central air nozzle and, thus, the temperature level at $h = 5$ mm drops with the typical delay of $\approx 180°$ for the mixing and transport. At $\alpha \approx 270°$–$360°$, the temperatures at $h = 5$ mm are relatively low and the heat release rate is at a minimum. Consequently, the swirl number reaches again the maximum and so does the inner recirculation, restarting the cycle.

4. Conclusions

A gas turbine model combustor for swirling CH_4/air diffusion flames at atmospheric pressure has been presented which offered full optical access for the application of various laser techniques. The characteristics of a flame with self-excited thermoacoustic oscillations of $f = 290$ Hz have been investigated using phase-resolved LDV, 2D LIF, laser Raman scattering, and chemiluminescence measurements. The instantaneous flame behavior was dominated by strong turbulent fluctuations, however, periodic variations became clearly visible by averaging the measured results at distinct phase angles.

The flame burnt lifted and fuel, air, and exhaust gas were partially premixed before ignition. The region of the main heat release was restricted to $h \approx 10$–20 mm and mixing and combustion were completed at $h \approx 25$ mm. The largest phase-dependent variations occurred in the inner and outer recirculation zones with a pronounced axial movement of the IRZ (large variation of u) and radial movement of the ORZ (large variation of v). During an oscillation cycle, the heat release rate changed by more than $\pm 20\%$, especially in the outer region of the flame ($r \approx 15$–30 mm). These variations of heat release were not well correlated with changes of the mixture fraction except for $r \geq 25$ mm at the periphery of the flame. The results revealed that the temperature of the unburned fuel/air mixtures at $h = 5$ mm, i.e., at the beginning of the flame zone varied in phase with the total heat release. This temperature variation was due to a periodic change of the transport of hot products from the recirculation zones to the inlet flow and hardly to flame reactions. The oscillations of the heat release rate were accompanied by periodic changes of the thermal expansion, swirl intensity, and pressure, which in turn influenced the flow field. The variations were correlated in such a way that temperature, heat release, and transport sustained a periodic oscillation at $f \approx 290$ Hz.

The experimental results yielded a rather complete characterization of the GT-like flame investigated and explained some basic mechanisms leading to the sustained thermoacoustic oscillations. Furthermore, the results represent a quantitative data base that can be used for the improvement of theoretical combustion models and the validation of numerical simulations.

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